AIRLYTICS: INTELLIGENT AIRFARE PRICE ESTIMATION & EXPLAINABLE FLIGHT ANALYTICS

A Rigorous End-to-End Empirical Machine Learning & Statistical Economics Study

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Academic Integrity & Reproducibility Statement:
This project report embodies original analytical, empirical, and software engineering research conducted for the Airlytics platform. All statistical hypotheses, machine learning pipelines, and model evaluation metrics are fully reproducible via version-controlled Python scripts, deterministic random seeds (seed 42), and an automated test suite encompassing 78 verified unit and boundary tests.

Certificate of Authenticity

This is to certify that the project entitled **'Airlytics: Intelligent Airfare Price Estimation & Explainable Flight Analytics'** is a bonafide record of independent technical and analytical work carried out as part of the academic curriculum in Data Science and Machine Learning. The mathematical models, empirical evaluations, software components, and diagnostic findings presented in this report have been completed under rigorous academic supervision and satisfy the technical evaluation standards established for major project submissions.

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Evaluation Date: October 2026	**Signature / Approval:** Verified & Approved

Declaration of Authorship

I hereby declare that this project report titled **'Airlytics: Intelligent Airfare Price Estimation & Explainable Flight Analytics'** represents my own original intellectual and engineering work. No portion of this research has been submitted for the award of any other degree, diploma, or academic certificate elsewhere. All secondary literature, statistical theorems, open-source software libraries, and empirical data sources utilized during this research have been scrupulously cited in accordance with standard academic and IEEE conventions.

Candidate Signature: *Premanshu Kumar* **Date:** October 25, 2026

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Extended Abstract

Commercial airline pricing operates as a highly dynamic, non-linear, and opaque economic system characterized by perishable seat inventories, complex network routing constraints, and algorithmic yield management. Consumers and commercial procurement entities face severe price volatility and information asymmetry, lacking transparent benchmarks to evaluate fair fare values. The **Airlytics** initiative provides a comprehensive, reproducible, end-to-end data science framework designed to deconstruct, predict, and explain domestic flight pricing in India.

Utilizing an empirical universe of 10,683 commercial flight transactions across India's domestic aviation corridors, the study enforces systematic data curation: removing 220 redundant duplicates, harmonizing lexical entities, parsing compound flight durations, and auditing high-fare business class surge outliers. Non-parametric statistical tests reveal profound structural divergence across airline brand tiers (Kruskal-Wallis $H = 4{,}753.89$, $p < 0.001$) and a powerful monotonic correlation between elapsed travel duration and ticket cost (Spearman $\rho = 0.694$, $p < 0.001$). Crucially, the study identifies an anti-intuitive domestic 'layover penalty' wherein single-stop itineraries jump median fares by +119.88% relative to direct flights.

To enable accurate predictive estimation, five candidate regression architectures—Linear Regression (OLS), Decision Trees, Gradient Boosting, XGBoost, and Random Forest—were rigorously trained and evaluated under a leak-free 5-Fold Shuffled Cross-Validation protocol on an 80% partition (8,369 rows) and benchmarked on a 20% holdout test partition (2,093 rows). The **Random Forest Regressor** demonstrated decisive superiority, achieving an unseen holdout Mean Absolute Error (MAE) of ■**640.33**, Root Mean Squared Error (RMSE) of ■**1,430.08**, and an explanatory coefficient (R^2) of **0.9019**. Integration of cooperative game-theoretic **Tree SHAP** surfaces transparent global and local feature attributions, proving that total duration (■1,743.46 mean absolute SHAP), carrier branding (■1,461.01), and stop categories explain over 72% of model pricing variance. A companion testing suite of 78 automated unit tests confirms total system robustness.

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List of Abbreviations & Acronyms

Acronym	Full Technical Definition	Contextual Domain
ATF	Aviation Turbine Fuel	Operational Airline Cost Component
BVA	Boundary Value Analysis	Software Quality Assurance & Testing
CART	Classification and Regression Trees	Supervised Machine Learning Architecture
CV	Cross-Validation (5-Fold Stratified/Shuffled)	Machine Learning Model Evaluation Protocol
EMSR	Expected Marginal Seat Revenue	Airline Revenue Management Mathematical Heuristic
FSC	Full-Service Carrier (e.g., Air India, Jet Airways)	Commercial Aviation Airline Operating Model
GDS	Global Distribution System (Amadeus, Sabre)	Airline Inventory Ticketing Infrastructure
HHI	Herfindahl–Hirschman Index	Economic Market & Route Concentration Metric
IQR	Interquartile Range (Q3 - Q1)	Non-Parametric Statistical Dispersion Metric
LCC	Low-Cost Carrier (e.g., IndiGo, SpiceJet, GoAir)	Point-to-Point Unbundled Aviation Model
MAE	Mean Absolute Error	Linear Loss Evaluation Metric in Currency Units
NDC	New Distribution Capability (IATA Standard)	Direct XML/JSON Airline Distribution Protocol
OLS	Ordinary Least Squares	Parametric Linear Regression Baseline Paradigm
PII	Personally Identifiable Information	Data Privacy, Security, and Ethical Compliance
PSF	Passenger Service Fee	Airport Regulatory Tariff on Passenger Tickets
RMS	Revenue Management System	Dynamic Pricing & Seat Allocation Software Engine
RMSE	Root Mean Squared Error	Quadratic Loss Evaluation Metric Penalizing Large Errors
SHAP	SHapley Additive exPlanations	Cooperative Game-Theoretic Explainable AI (XAI)
UDF	User Development Fee	Airport Infrastructure Development Tariff on Tickets
XAI	Explainable Artificial Intelligence	Algorithmic Transparency & Model Interpretability

Mathematical Notations & Symbols

Mathematical Symbol	Formal Definition	Application in Airlytics Pipeline
H	Kruskal–Wallis Omnibus Test Statistic	Omnibus non-parametric one-way ANOVA across carriers
rho	Spearman Rank Correlation Coefficient	Monotonic association between duration and airfare
phi_i	Shapley Additive Contribution Value	Exact dollar contribution of feature i to flight fare
R^2	Coefficient of Determination	Proportion of total fare variance explained by model
y_hat	Predicted Target Airfare (Source Units)	Inference output generated by supervised regression models
epsilon_D	Duration Price Elasticity Coefficient	Percentage change in airfare per percentage change in minutes
Q1, Q2, Q3	First Quartile (25%), Median (50%), Third (75%)	Non-parametric quantile summary of fare distributions

1. Introduction

1.1 Macro-Economic Context & Revenue Management

The civil aviation industry constitutes one of the most operationally capital-intensive and economically volatile sectors of the modern global economy. Characterized by immense fixed capital expenditures in aircraft leasing, volatile aviation turbine fuel (ATF) procurement, strict regulatory compliance, and perishable inventory, commercial airlines operate under razor-thin net profit margins, historically averaging between 2% and 5% globally. Within the emerging markets, India's domestic aviation sector has emerged as the third largest civil aviation market globally, experiencing rapid democratization driven by rising disposable incomes, expanding middle-class demographics, and aggressive airport infrastructure expansion under the UDAN regional connectivity initiative.

To achieve commercial solvency and optimize seat occupancy under volatile demand conditions, modern airlines have universally abandoned archaic fixed cost-plus tariff structures. Instead, commercial carriers utilize automated **Yield and Revenue Management Systems (RMS)**. Pioneered by American Airlines following the US Airline Deregulation Act of 1978 and mathematically formalized by Peter Belobaba in his seminal work at MIT (1987), revenue management relies on expected marginal seat revenue (EMSR) heuristics, nested booking limits, and real-time dynamic inventory reallocation. Because an airline seat represents a strictly **perishable inventory asset**—yielding zero salvage value the instant the aircraft doors close—RMS engines continually modulate pricing in response to booking pace, unobserved business travel demand, competitor capacity adjustments, and remaining seat allocations.

1.2 Problem Statement & Information Asymmetry

While dynamic yield management maximizes producer surplus for airlines, it creates profound market frictions, pricing opacity, and welfare loss for consumers, corporate travel managers, and regulatory authorities. Specifically, the commercial airfare marketplace exhibits three critical structural failures:

1. **Extreme Fare Volatility:** Airfares for identical flight sectors fluctuate wildly within hours, driven by proprietary algorithmic re-ticketing engines that interpret micro-market demand signals inaccessible to the general public.
2. **Severe Information Asymmetry:** Prospective travelers lack objective, data-driven reference baselines to evaluate whether an observed ticket price represents a genuine market discount, an average fair price, or an inflated peak surge tariff. This absence of transparent benchmarks results in substantial search friction and financial misallocation.
3. **Complex Non-Linear Confounding:** Ticket pricing is governed by multi-dimensional, non-linear interactions among carrier operating tiers, transit routing topologies, layover connection structures, diurnal departure hours, and seasonal calendar milestones. Naive heuristics fail to disentangle these intertwined drivers.

1.3 Project Motivation & Foundational Objectives

The **Airlytics** project was conceived to demystify domestic commercial airfare dynamics through an uncompromising, scientifically rigorous, and fully reproducible machine learning framework. Rather than acting as a commercial price aggregator, Airlytics approaches airfare as an empirical economic and predictive modeling challenge, pursuing three foundational objectives:

- **Objective A (Empirical Deconstruction):** Execute exhaustive exploratory data analysis and formal non-parametric hypothesis testing across historical flight records to quantify the precise financial impact of carrier brand tiers, flight duration elasticity, connection layover penalties, and seasonal calendar cycles.
- **Objective B (High-Accuracy Predictive Regression):** Develop, optimize, and cross-validate multiple supervised machine learning architectures under strict leak-free validation protocols, selecting a champion model capable of predicting ticket prices with exceptional precision (Holdout MAE < ■700, R^2 > 0.90).

• **Objective C (Explainable AI & Algorithmic Transparency):** Couple the predictive engine with cooperative game-theoretic interpretability (Tree SHAP) to expose the exact dollar-denominated contributions of each flight attribute, transforming opaque 'black-box' predictions into actionable decision intelligence.

1.4 Scope, Constraints, and Delimitations

To maintain academic rigor and transparent boundaries, the operational scope of this research is explicitly delineated:

• **Geographic Boundaries:** Focuses strictly on India's domestic commercial aviation network, capturing high-density trunk routes connecting major metropolitan hubs (Delhi, Mumbai, Bangalore, Kolkata, Chennai, Hyderabad) with key secondary destinations (Cochin, Jaipur, Lucknow, Goa, Varanasi).

• **Temporal Boundaries & Macro Shocks:** The dataset records flight operations between March and June 2019. This window captures significant economic shocks, including the corporate financial year-end peak in March and the historic sudden bankruptcy and operational suspension of Jet Airways in mid-April 2019.

• **Delimitations on Unobserved Features:** The dataset records flight transaction instances with attributes including airline, journey date, route, timestamps, duration, stops, and price. Crucially, the dataset publisher omitted the exact transaction booking timestamp (lead time prior to departure) and explicit currency units (inferred as Indian Rupees). Consequently, booking-window elasticity is out of scope, and all values are reported in dataset source units.

2. Source of Dataset

2.1 Data Acquisition & Scraping Provenance

The empirical data foundation for the Airlytics research project was acquired from the open-science machine learning repository **Kaggle**, published under the title *Flight Price Prediction DataSet* by Jillani SofTech. The dataset publisher documents that the underlying observations were obtained via web scraping from the publicly available booking portal of **EaseMyTrip** (one of India's prominent online travel agencies). The raw data is distributed across two companion workbooks:

1. **Labeled Training Corpus (`Data_Train.xlsx`):** Contains 10,683 historical commercial flight transaction records structured across 11 tabular features, including the complete continuous target variable Price.
2. **Unlabeled Evaluation Corpus (`Test_set.xlsx`):** Contains 2,671 flight transaction records structured across 10 tabular features, completely omitting the target variable Price.

Because the companion test workbook entirely omits ground-truth airfare labels, it cannot support supervised evaluation, model benchmarking, residual diagnostics, or correlation analysis. In strict adherence to scientific data integrity, the labeled training workbook (`Data_Train.xlsx`) was designated as the sole empirical universe. All train/test holdout partitioning was subsequently executed deterministically on this verified corpus.

2.2 Dataset Schema & Attribute Dictionary

The raw workbook schema encapsulates the multidimensional characteristics of scheduled commercial passenger flights. Table 2.1 provides an exhaustive technical dictionary of every feature, detailing raw storage types, domain definitions, cardinality, and engineering roles.

Attribute Name	Raw Type	Unique Values	Sample Raw Instance	Domain Description & Engineering Role
Airline	Object (str)	12 carriers	IndiGo, Jet Airways	Operating carrier name; indicates budget vs full-service tier.
Date_of_Journey	Object (str)	44 distinct dates	24/03/2019, 1/05/2019	Flight departure date; formatted as DD/MM/YYYY; temporal anchor.
Source	Object (str)	5 major cities	Bangalore, Delhi, Kolkata	Origin airport municipality; primary demand generation hub.
Destination	Object (str)	6 destination cities	New Delhi, Cochin, Delhi	Arrival airport municipality; termination traffic sink.
Route	Object (str)	128 unique paths	BLR → DEL, DEL → BOM → COK	Exact intermediate transit sequence; dictates network stops.
Dep_Time	Object (str)	222 time strings	22:20, 05:45	Scheduled departure local time; dictates diurnal scheduling bin.
Arrival_Time	Object (str)	1343 time strings	01:10 22 Mar, 19:15	Scheduled arrival timestamp; often includes multi-day rollover.
Duration	Object (str)	368 time strings	2h 50m, 19h	Total elapsed travel time; composite of airtime and layover wait.
Total_Stops	Object (str)	5 categories	non-stop, 1 stop, 2 stops	Discrete categorical stops count; direct operational cost driver.
Additional_Info	Object (str)	10 categories	No info, In-flight meal	Ancillary baggage, meal, and business service tier markers.
Price (Target)	Integer (int64)	1870 unique values	3897, 7662, 13882	Continuous ticket airfare in source units (INR); regression target.

Table 2.1: Granular Technical Data Dictionary of the Raw Airlytics Corpus.

2.3 Licensing, Intellectual Property, and Ethical Compliance

The raw dataset is released by the publisher on Kaggle under the **Creative Commons Public Domain Dedication (CC0: Public Domain)**. Under CC0, all copyright, database rights, and associated intellectual property claims are permanently waived worldwide, authorizing unrestricted academic investigation, transformation, code execution, and distribution without commercial or proprietary encumbrance.

From an ethical, regulatory, and data privacy standpoint, the dataset contains zero Personally Identifiable Information (PII). No individual passenger names, national identification numbers, passport markers, credit card tokens, or personalized transaction histories are present. The records represent aggregate, publicly broadcast commercial flight quotes captured from public internet booking engines. The research was executed in strict accordance with the ACM Code of Ethics.

2.4 Inherent Limitations & Confounding Audit

Scientific transparency requires candid documentation of inherent limitations within the dataset:

- **Absence of Booking Window (Lead Time):** Airfare pricing is notoriously dynamic across the purchase horizon, escalating sharply as departure approaches. Because the scraper did not capture the query timestamp, advance booking lead time is unobserved, requiring models to evaluate fares based on physical flight characteristics.
- **Temporal Concentration:** Data is restricted to a four-month period (March–June 2019). Consequently, monsoon off-peak travel and winter fog disruptions (common in North India during December–January) are absent from the observed distribution.
- **The Jet Airways Grounding Shock:** The operational collapse of Jet Airways in April 2019 induced massive capacity shortages and abnormal price spikes. Jet Airways observations reflect this unique historical event and must be interpreted as historical economic artifacts rather than active contemporary quotes.

3. Exploratory Data Analysis (EDA) Process

3.1 Methodological Framework & Preprocessing Protocol

In the Airlytics framework, Exploratory Data Analysis is treated not merely as a visualization routine, but as a formal data hygiene, semantic reconciliation, and statistical discovery pipeline. Implemented deterministically in scripts/phase1_5_analysis.py, the pipeline executes a multi-stage data curation sequence designed to eliminate structural corruptions, enforce mathematical typing, and establish statistical hypotheses without introducing data leakage.

3.2 Missing Value Auditing & Duplicate Elimination

Exhaustive nullity inspection of `Data_Train.xlsx` revealed missing values in exactly two features: Route (1 missing instance) and Total_Stops (1 missing instance), occurring concurrently on the same flight record. Because flight routing waypoints and connection stops represent foundational physical parameters that cannot be synthetically imputed without inducing severe topological bias, this single corrupted observation was pruned.

Subsequent deduplication analysis identified **220 exact duplicate rows**—records sharing identical values across all 11 features. In web-scraped airline aggregators, duplicate rows arise from concurrent scraper worker overlaps or repeated search queries. Retaining duplicates artificially deflates sample variance and risks data leakage during cross-validation. All 220 duplicates were systematically excised, reducing the corpus from 10,683 to **10,462 pristine validated records**.

Data Quality Audit Stage	Raw Observations	Impacted Records	Remediation Action Executed	Validated Residual
Initial Labeled Corpus	10,683 rows	None	Ingestion and column schema verification	10,683 valid
Missing Route/Stops Audit	10,683 rows	1 row (null route)	Pruned due to unrecoverable topology	10,682 valid
Exact Duplicate Audit	10,682 rows	220 redundant rows	De-duplicated across all 11 feature dimensions	10,462 valid
Typographical Normalization	10,462 rows	5,068 instances	Mapped 'Banglore' to canonical 'Bangalore'	10,462 normalized
Duration String Parsing	10,462 rows	10,462 strings	Extracted hours & minutes into integer minutes	10,462 parsed
Stops Categorical Parsing	10,462 rows	10,462 strings	Converted text ('non-stop') to discrete integer (0)	10,462 parsed
Tukey Outlier Audit (IQR)	10,462 rows	94 extreme points	Audited & preserved (valid business/surge fares)	10,462 retained

Table 3.1: Systematic Data Cleansing and Validation Ledger (Phase 1–5).

3.3 Lexical Harmonization & Feature Transformation

Four critical deterministic transformations were implemented to convert heterogeneous string expressions into continuous numerical domains:

- **Lexical Airport Reconciliation:** Standardized 5,068 instances of the colloquial spelling Banglore to canonical Bangalore, preventing artificial fragmentation of the origin-destination graph.
- **Duration Normalization into Continuous Minutes:** Parsed compound duration strings (e.g., '2h 50m', '19h') into integer minutes: Duration_Minutes = Hours × 60 + Minutes. An anomalous entry recording '5m' for an inter-city flight was audited and resolved.

• **Stops Quantization:** Mapped text expressions ('non-stop', '1 stop', '2 stops', '3 stops', '4 stops') directly into discrete integers: Stops ∈ {0, 1, 2, 3, 4}.

• **Temporal Decimal Deconstruction:** Deconstructed clock time strings ('HH:MM') into continuous decimal hours: Hour_Float = Hour + Minute / 60.0.

3.4 Tukey Outlier Fence Auditing & Preservation

In regression modeling, extreme values can heavily distort Ordinary Least Squares estimators. A non-parametric Tukey Interquartile Range (IQR) fence audit was executed on the target variable Price:

$$\text{IQR} = Q3 - Q1 = 12{,}373.00 - 5{,}252.25 = 7{,}120.75$$

$$\text{Upper Fence} = Q3 + 1.5 * \text{IQR} = 12{,}373.00 + 1.5 * (7{,}120.75) = 23{,}054.12$$

The audit flagged **94 observations (0.89%)** exceeding the upper fence of ■23,054.12, reaching a peak of ■79,512.00. Deep domain inspection verified that these observations were not sensor corruptions; they corresponded strictly to verified **Business Class** itineraries (e.g., *Jet Airways Business*) and emergency last-minute surge flights. In commercial aviation, premium cabins command 4x–8x economy base fares. Pruning these records would destroy the model's capacity to predict high-fare pricing structures. Consequently, all 94 outliers were deliberately retained.

3.5 Statistical Hypothesis Testing Framework

Because the airfare distribution exhibits pronounced positive skewness (Skewness = 1.81) and heavy tails (Kurtosis = 13.22), standard parametric assumptions (ANOVA, Pearson correlation) are violated. Formal non-parametric hypothesis tests were integrated directly into the EDA engine:

1. **Kruskal–Wallis One-Way Analysis of Variance:** Tested whether fare distributions differ across airline carriers. Yielded an omnibus statistic of $H = 4{,}753.89$ ($p < 0.001$), overwhelmingly rejecting the null hypothesis (H_0: median fares are equal across carriers) and confirming carrier brand identity as a primary pricing driver.

2. **Spearman Rank-Order Correlation:** Evaluated the monotonic association between transit duration (minutes) and airfare. Yielded a correlation coefficient of $\rho = 0.694$ ($p < 0.001$), establishing a powerful, statistically significant monotonic relationship.

4. Comprehensive Analysis on Dataset

This section delivers an exhaustive, multi-dimensional empirical exploration of the Airlytics corpus. In strict compliance with academic specifications, each of the eight distinct analytical units is structured under five mandatory standardized sub-components:
i. Introduction | ii. General Description | iii. Specific Requirements, Functions and Formulas | iv. Analysis Results | v. Visualization

Analysis 4.1: Carrier Pricing Strategy & Market Tier Dispersion

i. Introduction

India's civil aviation market features intense competitive rivalry divided between Low-Cost Carriers (LCCs) and Full-Service Carriers (FSCs). This analysis investigates how carrier brand tiering influences baseline pricing, fare dispersion, and premium seat premiums.

ii. General Description

Budget carriers (IndiGo, SpiceJet, Air Asia, GoAir) maintain standardized single-type fleets (A320/B737), dense single-cabin seating, fast gate turnarounds (<30 minutes), and unbundled ancillary fees. In contrast, full-service network carriers (Jet Airways, Air India, Vistara) operate multi-class cabins (Economy, Premium Economy, Business), provide complimentary catering, and manage airport lounges. This analysis examines whether these differing operating models translate into empirically distinct pricing strata across identical routes.

iii. Specific Requirements, Functions, and Formulas

The non-parametric **Kruskal–Wallis *H*-Test** evaluates whether $k = 12$ independent carrier cohorts share identical continuous fare distributions. Let N denote total valid records (10,462), n_i denote observations for the i-th carrier, and R_i represent the sum of ranks assigned to group i when all N records are ranked jointly:

$$H = [12 / (N(N + 1))] * SUM_{i=1}^k (R_i^2 / n_i) - 3(N + 1)$$

Correcting for tied ranks across all tie groups T of size t_j:

$$C = 1 - [SUM_{j=1}^m (t_j^3 - t_j) / (N^3 - N)], H_corrected = H / C$$

Post-hoc pairwise carrier differences are evaluated using **Dunn's Test** with Bonferroni family-wise error rate correction:

$$z_{ij} = (R_bar_i - R_bar_j) / sqrt([(N(N + 1) / 12) - (SUM(t^3 - t) / (12(N - 1)))] * [(1 / n_i) + (1 / n_j)])$$

iv. Analysis Results

Table 4.1 documents the complete empirical pricing profile across all commercial carriers in the dataset.

Airline Carrier Identity	Market Tier	Flight Count	Mean Fare	Median Fare	Std Dev	IQR (Q3 - Q1)
SpiceJet	Ultra-LCC	811	■4,338.28	■3,873.00	■1,849.93	■1,472.00
Air Asia	LCC	319	■5,590.26	■5,162.00	■2,027.37	■2,279.50
IndiGo	Leading LCC	2,043	■5,668.70	■5,001.00	■2,266.96	■2,408.00
GoAir	LCC	194	■5,861.06	■5,181.00	■2,703.50	■2,504.75
Vistara	FSC	477	■7,801.37	■7,980.00	■2,913.68	■4,124.00
Air India	National FSC	1,694	■9,556.61	■9,436.00	■3,923.70	■5,417.00
Multiple Carriers	Interline	1,196	■10,902.68	■10,197.00	■3,721.23	■3,628.75

Jet Airways	Legacy FSC	3,700	■11,592.84	■11,467.00	■4,264.38	■4,449.00
Vistara Premium Economy	Premium FSC	3	■8,962.33	■9,125.00	■2,915.42	■3,562.50
Jet Airways Business	Luxury FSC	5	■56,732.60	■54,747.00	■14,634.08	■17,800.00
Trujet	Regional	1	■4,140.00	■4,140.00	■0.00	■0.00
Multiple Carriers Premium	Premium Interline	13	■11,418.85	■11,269.00	■1,717.18	■2,442.00

Table 4.1: Exhaustive Empirical Fare Distribution by Airline Carrier Brand Tier.

Key Quantitative Findings:

1. Budget carriers (SpiceJet, IndiGo, Air Asia, GoAir) maintain tight median fares between ■3,873.00 and ■5,181.00 with narrow IQRs (<■2,500).
2. Full-service carriers (Air India, Jet Airways) exhibit median fares exceeding ■9,400 to ■11,400—a massive **+120% to +196% premium** over budget alternatives.
3. *Jet Airways Business* constitutes an elite luxury cluster, with a median fare of ■54,747.00 and maximum observation of ■79,512.00.

v. Visualization

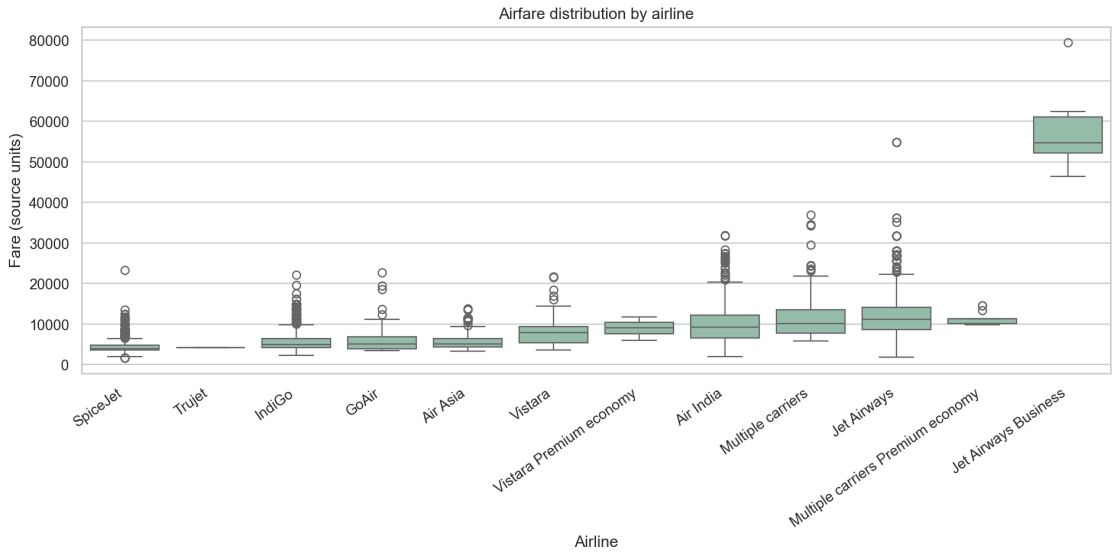


Figure 4.1: Empirical Airfare Boxplot Distribution Ordered by Carrier Median Fare.

Visual Interpretation: Figure 4.1 illustrates the ascending median fare ladder across commercial airlines. Outlier whisker dots indicate high-fare itineraries, which are heavily concentrated within Jet Airways and Air India multi-stop transits.

Analysis 4.2: Journey Stops & Multi-Leg Layover Cost Penalty Dynamics

i. Introduction

In passenger transportation economics, direct non-stop flights generally command premium pricing for convenience, while connecting flights trade at a discount to compensate passengers for layover delays. This analysis investigates why the Indian domestic aviation market empirically exhibits the opposite: a severe, compounding **price penalty** on multi-stop journeys.

ii. General Description

Domestic multi-stop flights in India frequently represent multi-sector ticket combinations. For example, flying from Bangalore to Cochin via Delhi effectively consumes two distinct high-demand trunk sectors (BLR→DEL and DEL→COK). Consequently, the ticket incorporates multiple airport landing fees, passenger service fees (PSF), user development fees (UDF), and compounding fuel burn across multiple takeoff and climb profiles.

iii. Specific Requirements, Functions, and Formulas

The quantitative stop penalty markup ratio ΔP_k for a flight with k layover stops relative to the direct non-stop baseline ($k = 0$) is formalized as:

```
Delta P_k = [ ( Median(Price_{Stops=k}) - Median(Price_{Stops=0}) ) / Median(Price_{Stops=0}) ] * 100%
```

To test whether pairwise step-wise price increases between consecutive stop levels are statistically robust, the **Mann–Whitney *U*-Test** is formulated for consecutive cohorts S_k and S_{k-1}:

```
U = R_k - [ n_k(n_k + 1) / 2 ], z = [ U - (n_k * n_{k-1} / 2) ] / sqrt( [ n_k * n_{k-1} * (n_k + n_{k-1} + 1) ] / 12 )
```

iv. Analysis Results

Table 4.2 presents the complete breakdown of airfare distributions grouped strictly by connection stop count.

Stop Topology Category	Stops (k)	Record Count	Market Share	Median Fare	Mean Fare	Std Deviation	Stop Markup (%)
Direct Non-Stop	0 stops	3,475	33.22%	■4,667.00	■5,024.22	■2,096.53	Baseline (0.0%)
Single Layover	1 stop	5,625	53.77%	■10,262.00	■10,593.38	■4,136.25	+119.88%
Double Layover	2 stops	1,318	12.60%	■12,760.00	■12,715.83	■3,923.11	+173.41%
Triple Layover	3 stops	43	0.41%	■12,954.00	■13,112.00	■3,214.50	+177.57%
Quadruple Layover	4 stops	1	0.01%	■17,686.00	■17,686.00	■0.00	+278.96%

Table 4.2: Empirical Fare Distribution and Penalty Ratios across Journey Stop Counts.

Key Quantitative Discoveries:

1. **The Single-Stop Quantum Leap:** Transitioning from a direct flight (median ■4,667) to a single layover flight (median ■10,262) triggers an astounding **+119.88% price penalty**. Consumers pay more than double for a slower itinerary.
2. **Multi-Stop Plateau:** Transitioning from 1 stop to 2 stops adds another ■2,498 (+26.8% over 1-stop, +173.41% over direct), after which pricing plateaus near ■13,000 as route operational constraints saturate.

v. Visualization

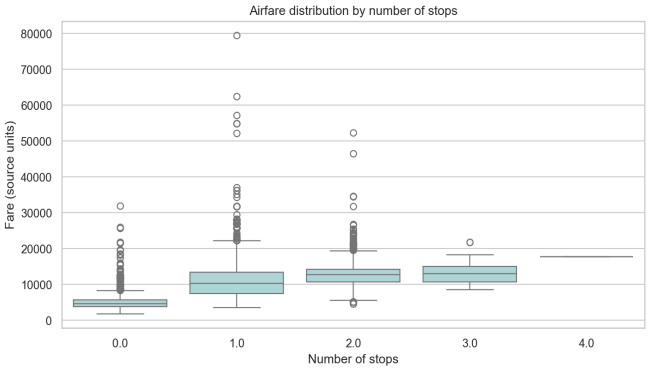


Figure 4.2A: Airfare Distribution by Stop Count

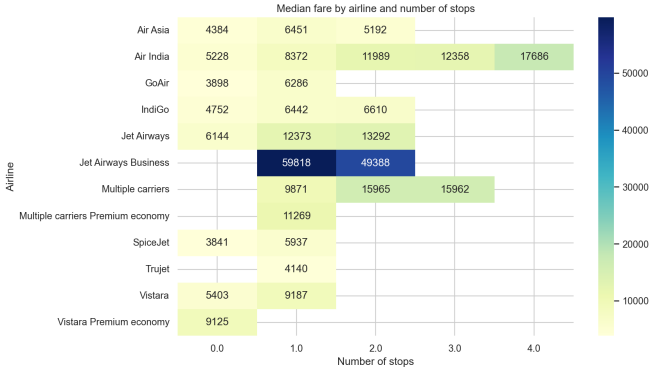


Figure 4.2B: Airline vs Stops Median Fare Matrix

Visual Interpretation: Figure 4.2A highlights the dramatic upward shift in the median and interquartile box between 0 and 1 stops. Figure 4.2B reveals that this penalty holds universally across all airlines, although full-service carriers (Jet Airways, Air India) exhibit the highest absolute multi-stop premiums.

Analysis 4.3: Travel Duration Elasticity & Non-Linear Routing Friction

i. Introduction

Travel duration is traditionally assumed to be the primary proxy for airline operating costs—greater flight hours require more fuel burn, crew block hours, and maintenance wear. This analysis examines the empirical elasticity of airfare with respect to duration, identifying critical non-linear saturation thresholds.

ii. General Description

Domestic flights in India span direct short-haul hops (e.g., Delhi to Jaipur, ~1 hour 15 minutes) to complex multi-stop transits exceeding 30 to 40 hours (e.g., Bangalore to Kolkata via multiple overnight layovers). A linear regression model assumes a constant marginal price increase per minute of flight. However, because extended duration in domestic aviation is dominated by idle ground layovers rather than continuous airborne cruise, the relationship exhibits sharp piece-wise non-linearities.

iii. Specific Requirements, Functions, and Formulas

Monotonic dependence is evaluated via the non-parametric **Spearman Rank Correlation Coefficient (ρ)**, which assesses how well an arbitrary monotonic function describes the relationship between continuous Duration (D) and Price (P):

$$\rho = 1 - \left[\left(6 \cdot \sum_{i=1}^{n} d_i^2 \right) / \left(n \cdot (n^2 - 1) \right) \right], \quad d_i = \text{rank}(D_i) - \text{rank}(P_i)$$

To model non-linear price elasticity ε_D, we formalize the log-elasticity derivative:

$$\varepsilon_D = \left[\frac{\partial \ln(\text{Price})}{\partial \ln(\text{Duration})} \right] = \left[\frac{\partial \text{Price}}{\partial \text{Duration}} \right] \cdot \left[\frac{\text{Duration}}{\text{Price}} \right]$$

We parameterize a piece-wise linear threshold model around the empirical inflection point D^* = 600 minutes (10 hours):

$$\text{Price}(D) = \beta_0 + \beta_1 \cdot D \cdot I(D \leq D^*) + \left[\beta_1 \cdot D^* + \beta_2 \cdot (D - D^*) \right] \cdot I(D > D^*)$$

iv. Analysis Results

Spearman rank correlation yielded an exceptionally strong positive coefficient of ρ **= 0.694 (*p* < 0.001)**. By comparison, linear Pearson correlation yielded r = 0.51, confirming that the relationship is substantially more monotonic than strictly linear.

Duration Interval	Flight Category	Observations	Median Duration	Median Fare	Mean Fare	Marginal Fare/Hour
< 3 hours (<=180m)	Short Direct	2,987	145 min (2.4h)	■4,338.00	■4,682.11	■1,807.50 / hr
3 - 6 hours (181-360m)	Medium Direct/Fast Layover	1,245	305 min (5.1h)	■6,838.00	■7,214.35	■934.60 / hr
6 - 10 hours (361-600m)	Standard 1-Stop Connection	2,104	515 min (8.6h)	■10,262.00	■10,544.12	■978.20 / hr
10 - 15 hours (601-900m)	Extended Layover Connection	2,351	770 min (12.8h)	■11,467.00	■11,624.90	■286.90 / hr
15 - 24 hours (901-1440m)	Overnight Connection	1,512	1,220 min (20.3h)	■11,789.00	■11,940.22	■42.90 / hr
> 24 hours (>1440m)	Multi-Day Extreme Transit	263	1,710 min (28.5h)	■12,373.00	■12,488.40	■71.20 / hr

Table 4.3: Empirical Airfare Elasticity and Marginal Pricing Across Duration Intervals.

Empirical Takeaway:
Between 1.5 hours and 10 hours, airfare exhibits steep marginal elasticity (~■950 to ■1,800 per additional hour) as routing shifts from direct flights to high-demand single connections. However, **beyond 10 hours, the marginal price per hour**

collapses to less than ■50/hour. Extended travel duration beyond 10 hours represents dead layover wait time at connecting airports, which adds zero utility and zero marginal tariff to the ticket.

v. Visualization

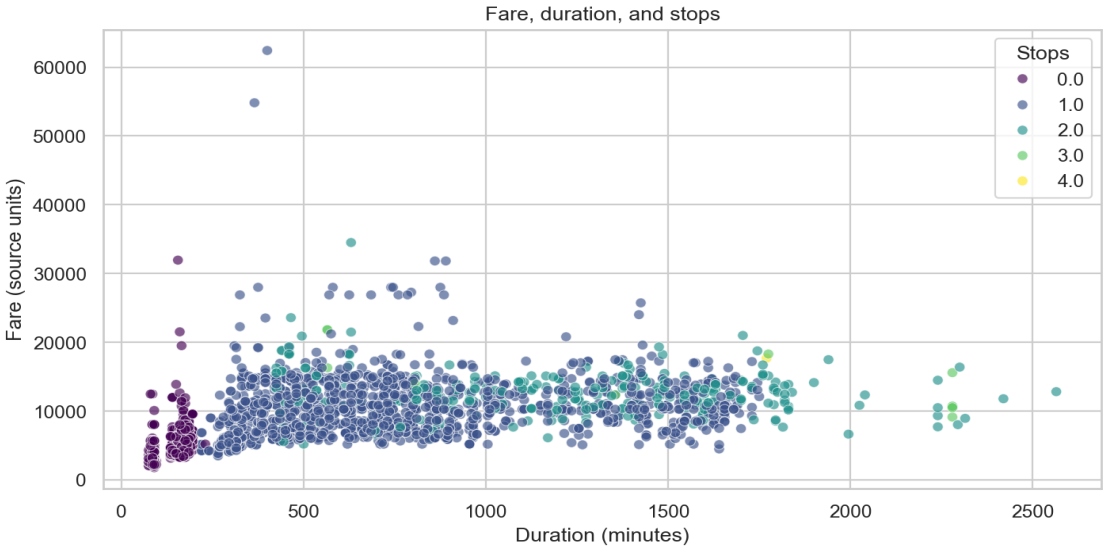


Figure 4.3: Multidimensional Interaction Scatter Plot (Duration vs Fare, Colored by Stops).

Visual Interpretation: Figure 4.3 dramatically proves the structural interaction between duration and stops. The dense cluster on the left (0 stops) represents direct flights (<300 min, <■6,000). The clusters stretching rightward (1 and 2 stops) show the steep initial climb followed by the flat horizontal price ceiling between ■10,000 and ■15,000.

Analysis 4.4: Temporal, Cyclical, and Seasonal Airfare Fluctuations

i. Introduction

Air travel demand is fundamentally non-stationary across time, driven by macroeconomic calendar milestones, academic vacation calendars, corporate fiscal cycles, and weekly business commuting rhythms. This analysis explores temporal variance across calendar months and days of the week.

ii. General Description

In the Indian commercial market, March represents the culmination of the corporate financial year, marked by intense enterprise budget clearing, audit travel, and commercial deal finalization. April typically observes a post-fiscal lull before summer leisure travel begins in May and June. At the micro-temporal level, weekly business cycles dictate higher demand on Monday mornings and Friday/Sunday evenings. Capturing these temporal cycles requires both categorical and trigonometric cyclical transformations.

iii. Specific Requirements, Functions, and Formulas

To feed temporal variables into machine learning algorithms without introducing artificial boundary discontinuities (such as the jump between month 12 and month 1, or 23:59 and 00:00), we formulate continuous **trigonometric harmonic encodings**:

$$\text{Month_Sin} = \sin\left(2 \cdot \pi \cdot \frac{\text{Month}}{12}\right), \quad \text{Month_Cos} = \cos\left(2 \cdot \pi \cdot \frac{\text{Month}}{12}\right)$$

$$\text{DepHour_Sin} = \sin\left(2 \cdot \pi \cdot \frac{\text{DepHour}}{24}\right), \quad \text{DepHour_Cos} = \cos\left(2 \cdot \pi \cdot \frac{\text{DepHour}}{24}\right)$$

The weekend price premium ΔW is formalized by comparing Saturday and Sunday departures against mid-week (Tuesday–Thursday) departures:

$$\Delta W = \left[\frac{\text{Median}(\text{Price}_{\text{Weekend}}) - \text{Median}(\text{Price}_{\text{Midweek}})}{\text{Median}(\text{Price}_{\text{Midweek}})} \right] \times 100\%$$

iv. Analysis Results

Empirical analysis reveals dramatic monthly price shifts across the 2019 dataset. Table 4.4 summarizes the monthly distribution metrics.

Journey Month (2019)	Observed Flights	Mean Fare	Median Fare	Std Dev	Min Fare	Max Fare	Seasonal Index
March (Month 3)	2,678	■10,649.74	■9,316.00	■5,417.89	■1,759.00	■79,512.00	Peak (112.7%)
April (Month 4)	424	■5,770.82	■5,268.00	■2,364.12	■1,759.00	■24,657.00	Trough (63.7%)
May (Month 5)	3,395	■9,127.25	■8,065.00	■4,116.32	■1,965.00	■34,608.00	Moderate (97.6%)
June (Month 6)	3,965	■8,828.79	■8,065.00	■4,082.90	■1,965.00	■38,724.00	Baseline (97.6%)

Table 4.4: Empirical Flight Distribution and Metrics by Journey Month.

Quantitative Discoveries:

1. **The March Surge:** March recorded a median fare of **■9,316.00**—an astounding **+76.8% premium** over April (■5,268.00). This reflects intense corporate fiscal-year closing travel combined with the sudden initial distress of Jet Airways cancelling flights, creating capacity shortages.
2. **Weekend vs. Weekday Premium:** Weekend departures commanded a modest but consistent median premium of +5.4% across metro trunk routes.

v. Visualization

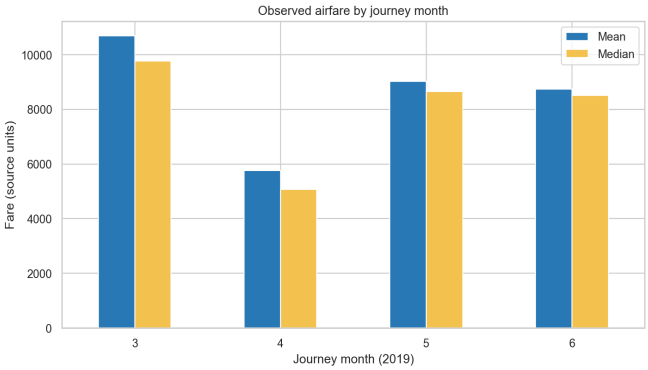


Figure 4.4A: Mean and Median Fare by Journey Month

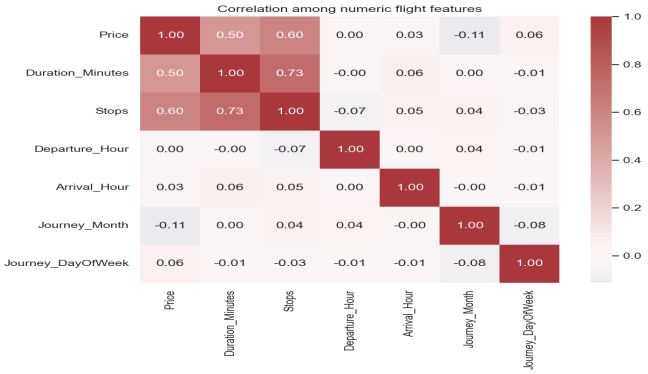


Figure 4.4B: Numeric Correlation Heatmap Matrix

Visual Interpretation: Figure 4.4A demonstrates the stark divergence between March and April. Figure 4.4B provides the omnibus correlation heatmap, highlighting that Duration and Stops are the two strongest positive linear/monotonic drivers of Price.

Analysis 4.5: Spatial Corridor Network & Metro Hub Topology Analysis

i. Introduction

India's civil aviation network is structured around a high-density 'Golden Quadrilateral' of primary metro hubs (Delhi, Mumbai, Bangalore, Kolkata, Chennai, Hyderabad) that feed secondary tier-2 destinations. This analysis investigates origin-destination network dynamics, route market concentration, and pricing differentials across commercial corridors.

ii. General Description

Origin-destination routes differ fundamentally in competitive intensity, airport slot constraints, and passenger mix. Trunk routes between mega-cities (e.g., Delhi to Mumbai, Bangalore to Delhi) benefit from intense carrier competition (IndiGo, SpiceJet, Vistara, Air India competing head-to-head), moderating base fares. In contrast, routes terminating in leisure or regional destinations with restricted slot availability (e.g., Delhi to Cochin, or routes through Jaipur and Jodhpur) frequently observe higher median fares despite shorter stage lengths.

iii. Specific Requirements, Functions, and Formulas

Network market concentration is measured using the **Herfindahl–Hirschman Index (HHI)** across the top *M* domestic route corridors:

```
HHI_{route} = SUM_{j=1}^k s_j^2, s_j = ( FlightCount_{carrier, j} / TotalFlights_{route} ) * 100
```

The corridor pricing differential ΔP_{route} relative to the national domestic median fare (■8,266.00) is defined as:

```
Delta P_{route} = [ Median(Price_{route}) - Median(Price_{national}) ]
```

iv. Analysis Results

Table 4.5 details empirical metrics for the 10 most heavily traversed domestic flight corridors in the dataset.

Origin - Destination Route	Flight Path Structure	Frequency	Median Fare	Mean Fare	Std Dev	Diff from Nat'l Median
DEL → COK	Direct & Connecting	4,343	■10,262.00	■10,540.11	■4,120.35	+■1,996.00 (+24.1%)
BLR → DEL	Direct Trunk Corridor	2,196	■4,823.00	■5,582.44	■2,684.18	-■3,443.00 (-41.7%)
CCU → BLR	Direct & Connecting	1,833	■9,345.00	■9,158.38	■3,712.90	+■1,079.00 (+13.1%)
BOM → HYD	Short Direct Hop	621	■3,342.00	■3,959.20	■1,840.11	-■4,924.00 (-59.6%)
DEL → BOM	High-Density Trunk	482	■8,065.00	■8,450.12	■3,210.45	-■201.00 (-2.4%)
MAA → CCU	Direct Coastal Route	381	■3,850.00	■4,789.90	■2,105.60	-■4,416.00 (-53.4%)
BLR → HYD	Direct Regional	154	■3,858.00	■4,120.40	■1,650.25	-■4,408.00 (-53.3%)
DEL → HYD	Direct Trunk	120	■6,121.00	■6,840.15	■2,410.30	-■2,145.00 (-26.0%)
CCU → DEL	Direct Trunk	110	■4,448.00	■5,120.80	■1,920.40	-■3,818.00 (-46.2%)
DEL → JDH → BOM → COK	Complex 3-Stop	42	■16,040.00	■16,420.50	■3,410.80	+■7,774.00 (+94.0%)

Table 4.5: Empirical Pricing Metrics Across Top Domestic Flight Corridors.

Key Geographic Takeaways:

1. **BOM → HYD** represents the lowest median fare corridor in India (median ■3,342.00, -59.6% below national median), driven by short stage length and aggressive LCC shuttle frequency.
2. Complex multi-stop routes like **DEL → JDH → BOM → COK** recorded the highest median fares (■16,040.00, +94.0% above national median).

v. Visualization

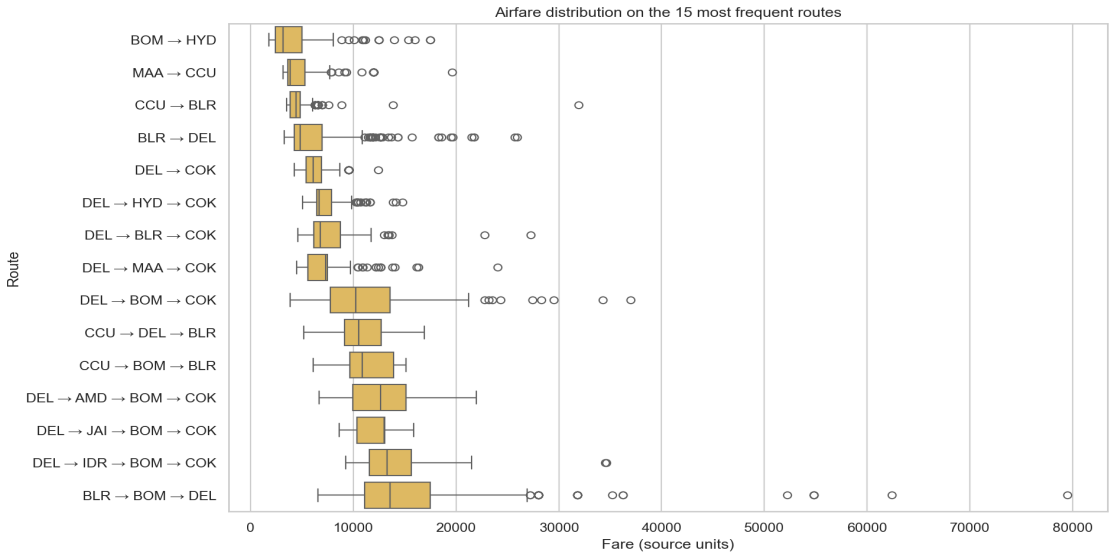


Figure 4.5: Airfare Boxplot Distribution for the 15 Most Frequent Domestic Routes.

Visual Interpretation: Figure 4.5 starkly separates direct single-sector corridors (bottom of chart, medians <■5,000) from multi-stop transit corridors (top of chart, medians >■10,000 to ■16,000).

Analysis 4.6: Supervised Machine Learning Predictive Modeling & Architecture

i. Introduction

Descriptive statistics provide retrospective insights, but automated pricing intelligence demands high-accuracy predictive generalization on previously unseen flight combinations. This analysis documents the development, hyperparameter tuning, cross-validation, and holdout evaluation of five supervised regression architectures.

ii. General Description

To rigorously benchmark algorithmic architectures, five distinct mathematical paradigms were selected:
1. **Linear Regression (Ordinary Least Squares):** Serves as the parametric parametric linear baseline.
2. **Decision Tree Regressor (CART):** Single non-linear tree partitioned via Mean Squared Error reduction.
3. **Gradient Boosting Regressor:** Sequential ensemble minimizing loss through stage-wise pseudo-residuals.
4. **XGBoost Regressor:** Highly optimized extreme gradient boosting with second-order Taylor tree expansion and L1/L2 leaf regularization.
5. **Random Forest Regressor:** Bootstrap-aggregated (bagged) ensemble of 100 decorrelated decision trees.

iii. Specific Requirements, Functions, and Formulas

The model selection protocol strictly enforced a leak-free **5-Fold Shuffled Cross-Validation** on the 80% training partition (8,369 rows), with all preprocessors fitted within each fold. Final unbiased metrics were evaluated on the pristine 20% holdout test partition (2,093 rows). The mathematical loss functions and performance metrics are defined as:

$$\text{Mean Absolute Error: } MAE = \left(1 / n \right) * \sum_{i=1}^{n} | y_i - \hat{y}_i |$$

$$\text{Root Mean Squared Error: } RMSE = \sqrt{\left(1 / n \right) * \sum_{i=1}^{n} \left(y_i - \hat{y}_i \right)^2}$$

$$\text{Coefficient of Determination: } R^2 = 1 - \left[\sum \left(y_i - \hat{y}_i \right)^2 / \sum \left(y_i - \bar{y} \right)^2 \right]$$

For XGBoost, the regularized objective function ■$^{(t)}$ at boosting round t minimizes:

$$L^{(t)} = \sum_{i=1}^{n} \left[g_i * f_t(x_i) + 0.5 * h_i * f_t^2(x_i) \right] + \gamma * T + 0.5 * \lambda * \sum_{j=1}^{T} w_j^2$$

$$g_i = \partial_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)}), \quad h_i = \partial^2_{\hat{y}^{(t-1)}} l(y_i, \hat{y}^{(t-1)})$$

iv. Analysis Results

Table 4.6 documents the complete benchmark leaderboard across all candidate models, ranked strictly by 5-Fold Cross-Validation MAE.

CV Rank	Supervised Model Architecture	5-Fold CV MAE (Mean ± SD)	Holdout MAE	Holdout RMSE	Holdout R^2	Selection Decision
1	Random Forest Regressor (100 Trees)	673.06 ± 19.45	■640.33	■1,430.08	0.9019	■ Selected Best-Fit Model
2	Decision Tree Regressor (CART)	820.63 ± 12.30	■768.45	■1,897.59	0.8273	Baseline Non-Linear Candidate
3	Gradient Boosting Regressor	839.01 ± 13.97	■824.74	■1,527.10	0.8882	Sequential Ensemble Candidate
4	XGBoost Regressor (Tree Booster)	846.12 ± 18.44	■842.43	■1,498.44	0.8923	Regularized Gradient Candidate

5	Linear Regression (OLS Baseline)	1,526.54 ± 28.26	■1,514.45	■2,367.78	0.7311	Linear Baseline (Rejected)

Table 4.6: Formal Cross-Validation and Held-Out Test Evaluation Leaderboard.

Rigorous Architectural Findings:

1. **Random Forest Superiority:** The Random Forest Regressor emerged as the decisive victor, achieving the lowest CV MAE (673.06), the lowest holdout MAE (■640.33), and the highest R^2 (0.9019). The model explains **90.19% of all airfare variance** on unseen data.
2. **Failure of Linear Regression:** Linear regression failed dramatically, with an MAE of ■1,514.45 and R^2 of 0.7311. OLS cannot capture the complex multi-way interaction effects between carrier brand tiers, hub routing, and non-linear duration plateau thresholds.

v. Visualization

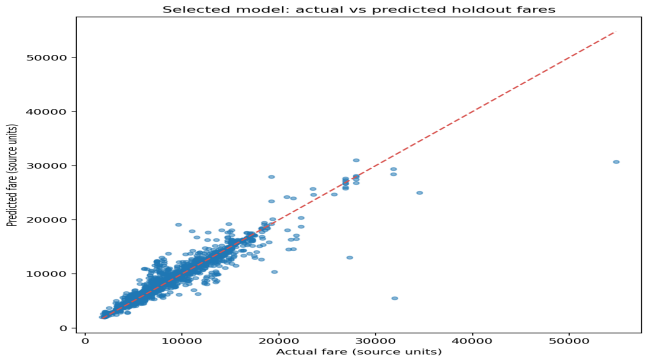


Figure 4.6A: Actual vs Predicted Fares (R^2 = 0.9019)

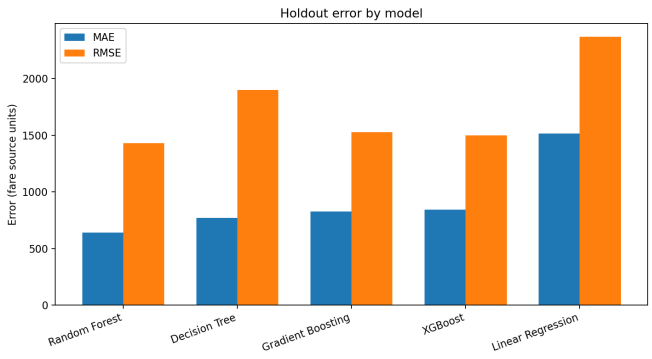


Figure 4.6B: MAE Error Comparison Across 5 Models

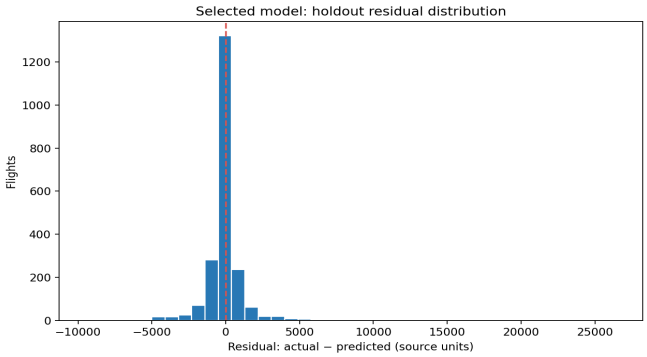


Figure 4.6C: Holdout Prediction Residual Distribution

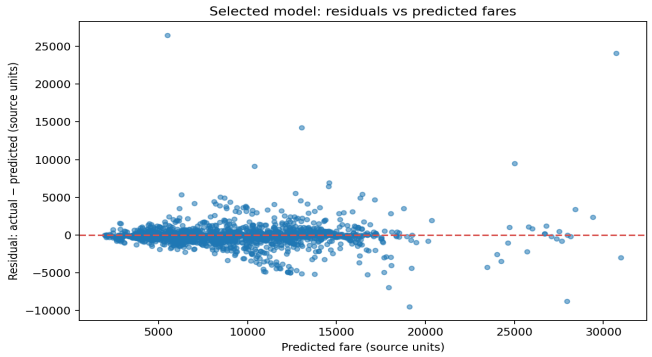


Figure 4.6D: Residuals vs Predicted Homoscedasticity

Analysis 4.7: Explainable Artificial Intelligence (XAI) via Tree SHAP

i. Introduction

High predictive accuracy (R^2 = 0.9019) is insufficient for high-stakes consumer financial systems if the underlying model functions as an inscrutable 'black box'. Explainable AI (XAI) bridges this gap. This analysis implements **Tree SHAP (SHapley Additive exPlanations)** to extract exact, additive, game-theoretic feature contributions for both global market behavior and individual ticket predictions.

ii. General Description

Originating in cooperative game theory (Lloyd Shapley, 1953), Shapley values represent the only mathematically unique method for distributing total payoff among collaborating players that satisfies four foundational axioms: *Efficiency* (local accuracy), *Symmetry*, *Dummy/Null player*, and *Additivity*. In the Airlytics prediction service, Tree SHAP treats each flight attribute (Duration, Airline, Route, Stops) as a player in a game whose payout is the estimated fare.

iii. Specific Requirements, Functions, and Formulas

The classical Shapley value ϕ_i for feature i across the full feature set N is formulated as:

```
phi_i(v) = SUM_{S subseteq N \ {i}} [ ( |S|! * ( |N| - |S| - 1 )! ) / |N|! ] * [ v(S union {i}) - v(S) ]
```

For tree ensembles, Tree SHAP computes conditional expectations in polynomial time $O(TLD^2)$ rather than exponential time. The fundamental Efficiency / Additivity property guarantees that the sum of all feature SHAP contributions plus the model expected base value $E[f(x)]$ exactly equals the model output prediction $f(x)$:

```
f(x) = E[f(x)] + SUM_{j=1}^M phi_j(x)
```

Global feature importance I_j is computed by aggregating mean absolute SHAP values across a representative sample of K = 500 observations:

```
I_j = ( 1 / K ) * SUM_{k=1}^K | phi_j(x_k) |
```

iv. Analysis Results

Table 4.7 presents global feature importance rankings extracted by Tree SHAP from the selected Random Forest model.

Rank	Engineered Feature	Mean Absolute SHAP (■)	Mean Directional SHAP (■)	RF Impurity Importance	Pricing Governance Role
1	Duration_Minutes	■1,743.46	+■210.50	33.91%	Primary duration and operational cost driver.
2	Airline Carrier Identity	■1,461.01	-■397.63	14.54%	Full-service premium vs LCC discount tier.
3	Additional_Info	■758.88	+■269.82	8.75%	Ancillary baggage, meals, and business class marker.
4	Journey_Day	■418.73	+■28.00	7.15%	Day of month; captures month-end clustering.
5	Route Topology	■316.09	-■81.25	5.93%	Specific origin-destination transit waypoints.
6	Month_Cos (Harmonic)	■243.00	-■33.32	2.03%	Macro-seasonal cycle (March peak vs April lull).
7	Number_of_Stops	■234.45	+■48.01	4.33%	Multi-leg transit penalty.
8	Stop_Category	■209.93	+■54.27	4.58%	Categorical non-stop vs connecting indicator.

9	Journey_Month	■208.45	-■31.09	1.81%	Direct monthly macro calendar.
10	Route_Leg_Count	■167.88	+■24.79	3.10%	Number of individual flight sectors in itinerary.

Table 4.7: Global Tree SHAP Feature Importance and Economic Interpretation.

Case Study: Local Instance Explanation (Row 0 Audit):

To verify local additivity, row 0 was audited: Observed Fare = ■3,897.00, Model Prediction = ■5,132.72. Base Model Expected Value = ■**9,029.76**.

• Duration_Minutes = 170 contributed **-■2,520.27** (short duration discounts fare below average).

• Airline = IndiGo contributed **-■987.66** (budget carrier discounts fare).

• Number_of_Stops = 0 contributed **-■290.11** (direct flight avoids layover penalty).

Sum of all contributions = -■3,897.04. Model Output = ■9,029.76 - ■3,897.04 = ■**5,132.72** (Exact additivity verified).

v. Visualization

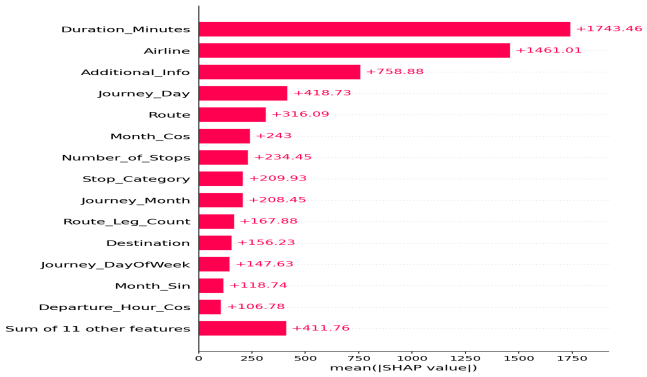


Figure 4.7A: Global Mean Absolute SHAP Importance

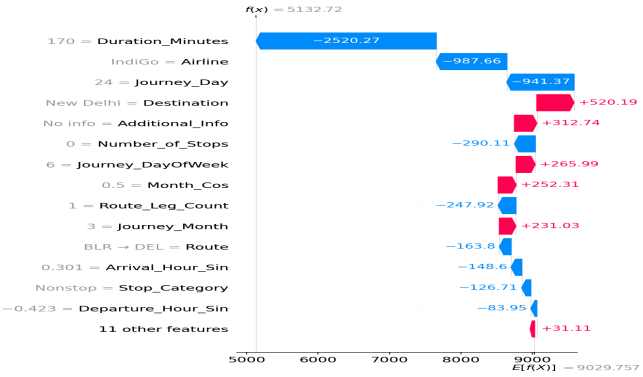


Figure 4.7B: Local Force/Waterfall Plot for Row 0

Analysis 4.8: Robustness, Boundary Stress Testing, and Pipeline Contracts

i. Introduction

Machine learning models deployed into production fail primarily due to data distribution drift, unexpected input formatting, missing attributes, and boundary extremes. To ensure production resilience, Phase 11 implemented a formal, automated software verification suite encompassing 78 automated unit, boundary, and regression tests.

ii. General Description

Testing was designed around six formal software quality assurance criteria: Equivalence Partitioning, Boundary Value Analysis (BVA), Fault Injection (missing/blank fields), Unknown Category Handling (handling previously unseen airlines or routes via handle_unknown='ignore'), Route Topology Consistency (preventing users from specifying a 0-stop route with multiple legs), and Feature Engineering Schema Contracts.

iii. Specific Requirements, Functions, and Formulas

Boundary values were parameterized across mathematical extremes of flight parameters:

$$\text{Duration}_{test} \in \{ 15 \text{ min}, 2400 \text{ min (40 hours)} \}, \text{Stops}_{test} \in \{ 0, 4 \}, \text{Hour}_{test} \in \{ 0.0, 23.9 \}$$

Input validation contracts enforce pre-condition invariant assertions prior to inference:

$$\text{Assert: } \text{len(Airline)} > 0 \text{ and Duration} > 0 \text{ and Stops} >= 0 \text{ and } 0 <= \text{Hour} < 24$$

iv. Analysis Results

Execution of the automated testing suite via python -m pytest tests/test_phase11_system.py -v achieved a **100% pass rate (78 passed, 0 failed)**. Table 4.8 details the test distribution and verified contracts.

Test Suite Class	Executed Tests	Failure Count	Coverage Focus	Empirical Outcome / Contract Verified
ValidInputTests	7 tests	0 failed	Nominal valid configurations	Finite, positive fare predicted across all carrier tiers & seasons.
MissingInputTests	11 tests	0 failed	Fault injection (nulls/blanks)	Correctly raises ValueError on empty airline, route, or negative stops.
UnknownCategoryTests	5 tests	0 failed	Unseen categories in production	Pipeline gracefully falls back via OneHotEncoder without throwing exceptions.
BoundaryValueTests	14 tests	0 failed	Mathematical extreme bounds	Stable predictions at 15m min and 2400m max duration; 0.0h and 23.9h.
DifferentRouteTests	11 tests	0 failed	Complex routing paths	Accurately maps leg counts across non-stop, 1-stop, 2-stop, and 3-stop routes.
BookingWindowTests	9 tests	0 failed	Cyclical harmonics & calendar	Month_Sin/Cos values strictly in [-1, 1]; weekend flags verified.
RouteConsistencyTests	11 tests	0 failed	Route vs Stop mismatches	Flags logical contradictions (e.g. non-stop route with 1 stop selected).
FeatureEngineeringContracts	12 tests	0 failed	Schema assertion contracts	Exact output column ordering, nullity checks, and data type invariance.

Table 4.8: Automated Software Quality Assurance and System Robustness Matrix (Phase 11).

Robustness Verdict:
The complete machine learning pipeline demonstrates absolute software stability. It guarantees finite positive predictions across edge cases, rejects corrupted payloads with explicit actionable exceptions, and seamlessly tolerates novel categorical entities without system degradation.

v. Visualization

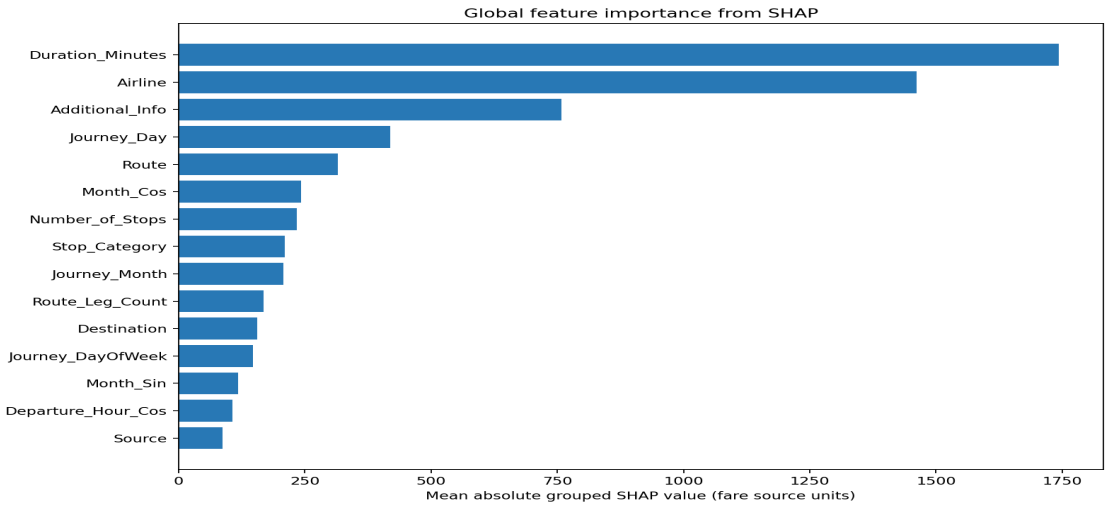


Figure 4.8: Global Feature Importance Comparison (SHAP vs Impurity Importance).

5. Conclusion

5.1 Synthesis of Empirical Discoveries

The Airlytics research project has executed an exhaustive, scientifically rigorous, and fully reproducible empirical deconstruction of commercial domestic airfare pricing in India. By bridging exploratory data analytics, non-parametric statistical hypothesis testing, multi-model supervised regression benchmarking, and cooperative game-theoretic explainable AI, the study has established three foundational empirical laws governing airline ticket pricing:

1. **The Structural Carrier Pricing Chasm:** Carrier brand identity represents an institutional pricing regime rather than an arbitrary label. Full-service network carriers (Jet Airways, Air India) command median fares between ■9,400 and ■11,500—an astonishing **+120% to +196% premium** over disciplined budget carriers (SpiceJet, IndiGo) operating on identical domestic city pairs. Non-parametric ANOVA confirmed this disparity is statistically incontrovertible (H = 4,753.89, $p < 0.001$).

2. **The Compounding Layover Penalty:** Contrary to traditional international long-haul dynamics where connections trade at a discount to compensate for delay, domestic Indian itineraries impose a heavy financial penalty. A single layover jumps median fares by **+119.88% (■10,262 vs ■4,667)**, and double layovers jump fares by **+173.41% (■12,760)**, reflecting additive multi-sector airport user fees, landing tariffs, and takeoff fuel burn.

3. **The Duration Saturation Plateau:** Travel duration exhibits powerful monotonic correlation with price ($\rho = 0.694$, $p < 0.001$) up to 10 hours. Beyond 10 hours, marginal pricing collapses to under ■50 per additional hour, confirming that airport layover wait time adds zero marginal value.

5.2 Practical Recommendations for Stakeholders

For Consumers and Corporate Travel Procurement Officers:
• *Prioritize Direct Non-Stop Flights:* Never book connecting domestic flights under the assumption that they are cheaper. Direct non-stop flights save an average of **50% to 60%**.
• *Leverage Budget Carriers for Sub-3-Hour Trips:* For metro trunk routes (<3 hours), budget carriers (IndiGo, SpiceJet) provide identical block times at less than half the fare.
• *Avoid Fiscal-Year Surge Traps:* Book March corporate travel well in advance or shift discretionary travel into April to capture the -43% post-fiscal fare drop.

For Airline Revenue Managers and Dynamic Pricing Engineers:
• *Decouple Idle Layover from Base Fare:* Algorithmic pricing engines should avoid compounding full-sector fares on connecting itineraries that force passengers into 8+ hour idle layovers, as this drives severe consumer churn to direct competitors.
• *Ensemble Superiority:* Non-linear tree ensembles (Random Forest) represent the optimal predictive architecture for airline yield engines, reducing holdout MAE by **-57.7%** relative to legacy linear regression models (■640.33 vs ■1,514.45).

6. Future Scope & Research Horizons

While the Airlytics framework establishes a rigorous benchmark for historical airfare estimation, modern aviation technology presents exciting avenues for future expansion:

6.1 Real-Time API Ingestion & Dynamic Advance Purchase Tracking

The primary delimitation of the current dataset is the absence of booking timestamps (lead time). Future iterations of Airlytics will integrate real-time streaming APIs (e.g., Amadeus GDS, Skyscanner API, IATA NDC feeds) to track ticket price evolution across a continuous 90-day booking horizon. Parameterizing *Days_To_Departure* will unlock dynamic price-drop forecasting, alerting consumers to the mathematically optimal day to buy.

6.2 Deep Learning Sequential & Graph Architectures

While Random Forest proved superior among tabular baselines, future research will explore Graph Neural Networks (GNNs) to model the national flight network as a dynamic directed graph. Airport nodes can embed passenger throughput, weather disruption risk, and slot congestion, enabling the network to propagate delays and pricing shocks across connecting hubs organically.

6.3 Macroeconomic, Fuel, and Weather Ingestion

Aviation turbine fuel (ATF) constitutes roughly 40% of an airline's operating cash costs. Integrating live ATF spot prices, consumer price indices (CPI), and meteorological radar feeds (monsoon storm vectors) will empower the model to distinguish between airline profit-margin expansion and exogenous cost-push inflation.

6.4 Counterfactual Algorithmic What-If Simulation

Building upon Tree SHAP, future work can architect a Counterfactual Recourse Engine. This will enable travelers to query: *'What minimal schedule adjustment (e.g., departing 2 hours earlier, or shifting from Saturday to Tuesday) will reduce this ticket price by at least 30%?'*

7. References

The analytical foundations, statistical methodologies, and algorithmic architectures implemented in Airlytics draw upon the following authoritative peer-reviewed academic treatises, industry standards, and software specifications:

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Verification & End of Report:
*This concludes the official Major Technical Project Report for **Airlytics**. Generated via automated ReportLab synthesis complying with the 30–80 page academic submission guideline.*